

Paper 3, Theorem 2.1: Statement and Interface Amendment

Internal research note for the **ShenWork** formalization

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Abstract

Theorem 2.1(1) of Chen–Ruau–Shen, Part II, claims uniform persistence under the sole condition $m \geq 1$, but its proof treats only $a = b = 0$ and $a, b > 0$. The omitted regime $a = 0 < b$ contains a spatially constant, positive, bounded, global solution that decays to zero, so the printed statement is false. The correct parameter guard is the disjunction actually used in the proof. A second issue concerns the formal interface for part (4): the paper’s mass is attached to the initial datum, whereas the first Lean version read the freely stored value $u(0)$ of a positive-time trajectory. This note gives the exact counterexamples, the corrected statements, and the corresponding axiom-clean Lean capstones.

1 Source checked

The source examined is the original submission, not a repository paraphrase:

Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, II: Persistence and stabilization*, arXiv:2604.02599v1, 3 April 2026.

Theorem 2.1 is titled “Uniform persistence.” Section 4.1 proves part (1), and Section 4.4 proves part (4). The two corrections below have different status: the first is a genuine parameter omission in the printed theorem; the second repairs the representation of the paper’s initial mass in the Lean trajectory API.

2 The omitted pure-decay regime in part (1)

The printed part (1) assumes only $m \geq 1$ and asserts, for every globally defined, bounded, positive solution,

$$\liminf_{t \rightarrow \infty} \inf_{x \in \Omega} u(t, x) > 0.$$

Section 4.1 nevertheless divides its proof into exactly two cases:

1. $a = b = 0$;
2. $a > 0$ and $b > 0$.

It never treats the allowed parameter regime $a = 0 < b$, and in that regime the conclusion is false.

2.1 Concrete counterexample

On any bounded Neumann domain, let $c > 0$, $a = 0$, and $b > 0$, and define the spatially constant functions

$$u(t, x) = (c^{-\alpha} + \alpha b t)^{-1/\alpha}, \quad v(t, x) = \frac{\nu}{\mu} u(t, x)^\gamma.$$

All spatial derivatives vanish. Hence the diffusion, chemotaxis-divergence, and boundary terms vanish, the elliptic equation holds identically, and the parabolic equation becomes

$$u_t = -b u^{1+\alpha}.$$

The solution is classical, strictly positive at every finite positive time, global, and bounded by c . However,

$$\lim_{t \rightarrow \infty} \inf_{x \in \Omega} u(t, x) = 0.$$

The Lean counterexample specializes to $\alpha = \gamma = m = \mu = \nu = b = 1$ and $a = \chi_0 = 0$, so that on positive time

$$u(t, x) = v(t, x) = \frac{1}{1+t}.$$

2.2 Corrected part (1)

The paper-faithful correction is the following.

Assume $m \geq 1$ and either $a = b = 0$ or $a > 0$, $b > 0$. Then every globally defined, bounded, positive solution has a strictly positive asymptotic spatial lower bound for u , and the displayed elliptic lower bound for v follows.

The remaining nonnegative-parameter regime $a > 0$, $b = 0$ does not produce a decaying counterexample. Integrating the equation and using both Neumann conditions gives

$$M'(t) = aM(t), \quad M(t) := \int_{\Omega} u(t, x) dx.$$

A positive solution therefore has exponentially growing mass and cannot be globally bounded. The persistence implication is vacuous in that regime. For fidelity, the recommended statement records the two cases proved in Section 4.1 instead of silently adding a vacuous third branch.

3 The positive-time mass interface in part (4)

Part (4) is a minimal-model statement ($a = b = 0$, $m = 1$). The paper starts from an initial datum u_0 , constructs its solution, and imposes the mass condition on u_0 . This is unambiguous because $u(t) \rightarrow u_0$ as $t \rightarrow 0^+$ and mass is conserved.

The first Lean statement instead used

$$\text{HasInitialMass}(D, u, u_*) \quad :\Longleftrightarrow \quad \int_D u(0) = |D|u_*.$$

But `IsPaper2GlobalClassicalSolution` constrains the trajectory only at strictly positive times. Its stored value $u(0)$ is free. Replacing only that zero-time slice preserves the full positive-time classical orbit, positivity, and boundedness, while changing `HasInitialMass` arbitrarily. Thus the old Lean version of part (4) is false for every admissible constants package.

The correct formal interface is the conserved physical mass

$$\forall t > 0, \quad \int_D u(t) = |D|u_*.$$

Equivalently, one may quantify over u_0 , require an actual initial-trace relation, and impose the mass condition on u_0 . The positive-time version is the shorter interface for the persistence proof and states exactly the invariant it consumes. This is a Lean-interface correction, not a claim that the paper’s mathematical part (4) is false.

4 Lean realization

The concrete interval counterexample to part (1) is

`ShenWork.Paper3. not_Theorem_2_1_part1_intervalDomain_pureDecay`

in `ShenWork/Paper3/IntervalDomainPersistencePart1StatementObstruction.lean`. It constructs the genuine interval-domain trajectory, proves global classical solvability and boundedness, computes the lower-envelope $\liminf$ as zero, and refutes the unguarded statement. The same file defines `Theorem_2_1_part1_corrected` with the Section 4.1 parameter split.

For part (4),

`ShenWork.Paper3. not_intervalDomain_Theorem_2_1_part4_anyConstants`

in `ShenWork/Paper3/IntervalDomainPersistenceMinimalMassInterfaceObstruction.lean` proves failure of the zero-slice formulation for every `Paper3Constants` package. The corrected capstone is

`ShenWork.Paper3. Theorem_2_1_part4_intervalDomainM_physicalMass_proven`

in `ShenWork/Paper3/IntervalDomainPersistenceMinimalPhysicalMass.lean`. It uses the paper-faithful u^m domain, conserved positive-time mass, the proved eventual upper bound, and the quantitative Neumann-resolver mass gap.

All three capstones print only the standard Lean/Mathlib axioms `propext`, `Classical.choice`, and `Quot.sound`.

5 Recommended amendment

1. Amend Theorem 2.1(1) by adding $(a = b = 0) \vee (a > 0 \text{ and } b > 0)$.
2. Retain the mathematical mass hypothesis of Theorem 2.1(4), but in Lean attach it to the initial datum plus initial trace, or use the equivalent conserved positive-time mass.
3. Do not use an axiom-clean proof as a substitute for semantic validation: both failures above are themselves provable without any nonstandard axiom.